

Visible upconversion in $\text{Er}^{3+}/\text{Yb}^{3+}$ co-doped LaAlO_3 phosphorsVijay Singh ^{a,*}, V.K. Rai ^b, N. Singh ^a, M.S. Pathak ^a, M. Rathaiah ^c, V. Venkatramu ^c, Rahul V. Patel ^d, Pramod K. Singh ^e, S.J. Dhoble ^f^a Department of Chemical Engineering, Konkuk University, Seoul 143 701, Republic of Korea^b Department of Applied Physics, Indian School of Mines, Dhanbad, 826 004, Jharkhand, India^c Department of Physics, Yogi Vemana University, Kadapa 516 003, India^d Department of Food Science and Biotechnology, Dongguk University-Seoul, Ilsandong-gu, Goyang-si, Gyeonggi-do, 410 820, Republic of Korea^e Materials Research Laboratory, Sharda University, Greater Noida 201 310, India^f Department of Physics, RTM Nagpur University, Nagpur 440 033, India

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ABSTRACT

The Er^{3+} doped and $\text{Er}^{3+}/\text{Yb}^{3+}$ co-doped LaAlO_3 phosphors have been synthesized by the combustion method and characterized their structural, morphological, elemental, vibrational and optical properties. The optical absorption and upconversion properties of the synthesized phosphors have been studied. Upon co-doping Yb^{3+} ions into $\text{Er}^{3+}:\text{LaAlO}_3$, the blue, green and red upconversion emissions of Er^{3+} ions have been enhanced about ... ~20, ~46 and ~22 times, under 978 nm laser excitation. The observed upconversion emissions could be due to excited state absorption in $\text{Er}^{3+}:\text{LaAlO}_3$, whereas energy transfer is dominant mechanism in $\text{Er}^{3+}/\text{Yb}^{3+}:\text{LaAlO}_3$ phosphors. The tuning in the color emitted from the synthesized phosphors towards the green region has been found due to incorporation of the Yb^{3+} ions. With increase in the pump power, the color emitted from the co-doped phosphor is not tuned significantly, showing its applicability in making the green display devices.

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1. Introduction

Trivalent lanthanide (Ln^{3+}) doped inorganic materials have attracted great attention due to their potential applications in optical devices such as optical fiber amplifiers, light converters, sensors, solid state lasers, and displays [1–4]. Among different families of Ln^{3+} -doped crystalline materials, oxide crystals are of great interest due to their unique optical properties, such as long fluorescent lifetime, large Stokes shift, their high chemical durability and thermal stability [5]. In particular, perovskite type oxides of general formula ABO_3 (the size of A cation is larger than B) is interesting to study. Lanthanum aluminate (LaAlO_3) with a perovskite-type structure is important one for many applications such as substrates for superconductors, magnetic and ferro-magnetic thin films, and luminescent host materials [6,7]. This is basically due to its special properties such as good thermal stability with high melting point (2180 °C) which can minimize the interfacial dislocations [8], good dielectric properties, high relative permittivity ($\epsilon_r = 23$), high quality factor (6800) and very small temperature coefficient of resonant frequency (-44 ppm/K) [9]. For instance, the LaAlO_3 is considered as a typical dielectric material with band gap energy ~5–6 eV [10], can be used as a host material for full-color phosphor applications [11].

Near-infrared (NIR) to visible light conversion in Ln^{3+} ion doped materials has great importance not only to understand the mechanisms of interaction between the Ln^{3+} ions in different hosts but also for their application in upconversion lasers, bio-labels, and bio-assays [12,13]. Among Ln^{3+} ions, the Er^{3+} is a promising ion for upconversion applications since it provides ladder type intermediate levels with long lifetimes, which are easily populated with low cost diode lasers. There has been a growing interest in upconverting phosphor compounds, which emit visible radiation when excited with infrared light. In fact, upconversion laser emissions in infrared and visible range have been achieved in several Er^{3+} doped phosphor materials [14,15].

In recent years, LaAlO_3 material has been well synthesized and characterized [7,16–20]. Liu et al. [7,16] reported the synthesis of Ln^{3+} doped LaAlO_3 ($\text{Ln}^{3+} = \text{Sm}, \text{Tb}$ and Tm) powders using a pechini-type sol-gel process and investigated their photoluminescence and cathodoluminescence properties. Mao et al. [17] proposed $\text{LaAlO}_3:\text{Eu}$ full color phosphor which could be great potential for application to white light-emitting devices. Thermally stimulated luminescence and electron spin resonance investigation on gamma irradiated $\text{LaAlO}_3:\text{Eu}^{3+}$ phosphor was reported by Singh et al. [18]. Fu and Liu [19] investigated photoluminescence and X-ray luminescence properties of the $\text{LaAlO}_3:\text{Eu}^{3+}$ nanophosphors. Dong et al. [20] synthesized Eu^{3+} doped LaAlO_3 phosphors through sol-gel process and investigated luminescent properties. Our literature survey shows as of now limited information is available on Ln^{3+} doped LaAlO_3 phosphors and in particular, no investigations on Er^{3+} doped and $\text{Er}^{3+}/\text{Yb}^{3+}$ co-doped LaAlO_3 phosphors.

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Hence, there is need to explore the infrared to visible energy conversion in Er^{3+} doped and $\text{Er}^{3+}/\text{Yb}^{3+}$ co-doped LaAlO_3 phosphors.

The Er^{3+} doped and $\text{Er}^{3+}/\text{Yb}^{3+}$ co-doped LaAlO_3 phosphors were prepared by combustion route. The advantages of this synthesis method are possibility to prepare complex composition, good homogeneity through mixing at the molecular level, safe, fast and required low temperature. In recent years, authors have shown that, by combustion synthesis, variety of complex oxides could be synthesized relatively at low temperatures [21,22]. A similar method was employed to produce $\text{Er}^{3+}:\text{LaAlO}_3$, co-doped with Yb^{3+} phosphor powders, in which urea has been employed as a fuel. The synthesized samples were characterized by X-ray diffraction (XRD), scanning electron microscopy (SEM) and energy dispersive analysis of X-ray (EDX), FTIR spectroscopic techniques for their structural, morphological and vibrational properties. The optical and upconversion properties have been studied by using diffuse reflectance and emission spectroscopy.

2. Experimental

$\text{LaAlO}_3:\text{Er}_{(0.03)}$ and $\text{LaAlO}_3:\text{Er}_{(0.03)},\text{Yb}_{(0.05)}$ were prepared by the combustion of $\text{La}(\text{NO}_3)_3 \cdot 6\text{H}_2\text{O}$, $\text{Al}(\text{NO}_3)_3 \cdot 9\text{H}_2\text{O}$, $\text{Er}(\text{NO}_3)_3 \cdot 5\text{H}_2\text{O}$, $\text{Yb}(\text{NO}_3)_3 \cdot 5\text{H}_2\text{O}$ and urea (NH_2CONH_2) as precursor materials. The optimal oxidizer:fuel ratio where the fuel and nitrates react completely was determined to be the stoichiometric composition. To perform the combustion reaction the appropriate materials were placed in a glass beaker of 100 ml capacity and mixed with de-ionized water. The resulting solution was allowed to react at 80°C for 20 min. to obtain a homogenous solution. Then the solution was introduced into a muffle furnace preheated to 550°C . After the reaction (which lasts about 5 min), the materials were crushed with a mortar and pestle and placed in 50 ml alumina crucibles to be heat-treated to 900°C for 4 h. The resulting powder was used for further characterization.

X-ray diffraction data were recorded on an X'Pert Pro Diffractometer (Panalytical) using $\text{CuK}\alpha_1$ radiation ($\lambda = 1.5406 \text{ \AA}$) in the 2θ range of 10° to 80° . Morphology was determined using a scanning electron microscope (SEM, Hitachi S-3200N, Japan). Their chemical compositions were analyzed using the energy dispersive X-ray (EDX) attached to the SEM. FT-IR spectra were taken using a Perkin-Elmer RX1 instrument in the range from 4000 to 400 cm^{-1} . The UV-Vis absorption spectra of the samples were measured in diffuse reflectance mode using a Cary 6000i UV-Vis-NIR spectrophotometer equipped with an integrating sphere. The upconversion emission spectra of powder samples were measured with a Fluorolog 3–22 spectrometer (Jobin Yvon) combined with a continuous wave 978 nm laser diode (LYPE 30-SGWL978-F400). The powder samples inside the spectrometer were excited at 978 nm laser via an optical fiber. All the measurements were carried out at room temperature.

3. Results and Discussion

3.1. Powder X-ray Diffraction: Structural Studies

The structure and phase purity of combustion derived $\text{Er}^{3+}:\text{LaAlO}_3$ and $\text{Er}^{3+}/\text{Yb}^{3+}:\text{LaAlO}_3$ phosphors are studied by XRD. The measured XRD pattern of (a) $\text{Er}^{3+}:\text{LaAlO}_3$ and (b) $\text{Er}^{3+}/\text{Yb}^{3+}:\text{LaAlO}_3$ phosphors are shown in Fig. 1. The perovskite structure with the crystalline planes of (012), (110), (202), (024), (122), (300), (220) and (312) are matched with that of pure LaAlO_3 (JCPDS no. 082-0478). It was observed that LaAlO_3 has a pure rhombohedral phase having centro-symmetric space group $R\bar{3}c$ (No. 167) with perovskite structure. As the ionic radii of Er^{3+} ($\sim 88 \text{ pm}$) and Yb^{3+} ($\sim 85 \text{ pm}$) ions are similar to the ionic radii of La^{3+} ions ($\sim 103 \text{ pm}$). The $\text{Er}^{3+}/\text{Yb}^{3+}$ ions occupy the La^{3+} ion crystallographic site with D_3 point symmetry [19]. There is no additional signature in the XRD pattern of the synthesized LaAlO_3 phosphors [Fig. 1 (a) & (b)] which imply that the dopants were incorporated into the structure of LaAlO_3 and its single phase formation. The

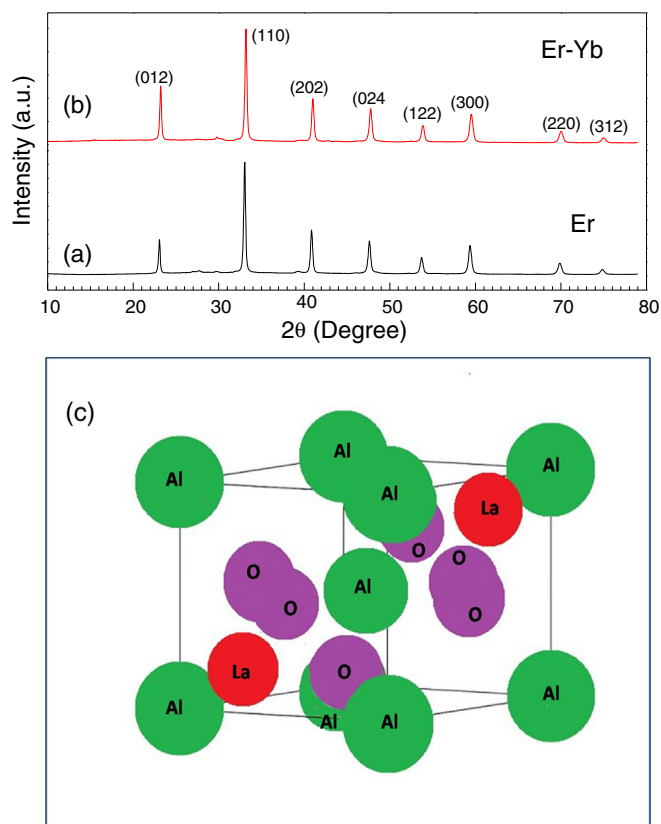


Fig. 1. Powder XRD pattern of (a) $\text{Er}^{3+}:\text{LaAlO}_3$ (b) $\text{Er}^{3+}/\text{Yb}^{3+}:\text{LaAlO}_3$ phosphors and (c) The crystal structure of LaAlO_3 unit cell.

crystal structure of LaAlO_3 unit cell is given in Fig. 1 (c). The XRD analysis indicates that the dopants do not participate in overall crystal formation except replacing/sitting between few atoms.

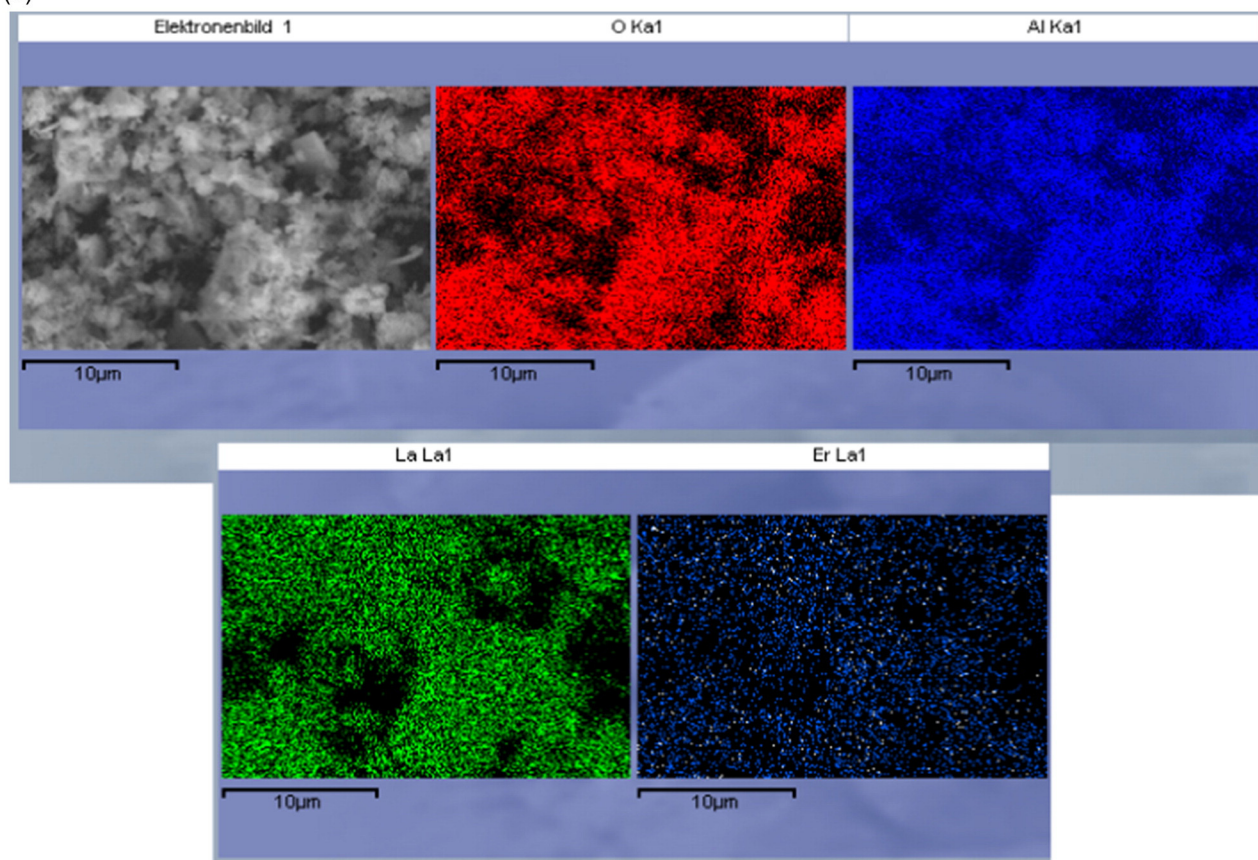
3.2. SEM: Morphological and Elemental Analysis

The morphology and elemental analysis of $\text{Er}^{3+}:\text{LaAlO}_3$ and $\text{Er}^{3+}/\text{Yb}^{3+}:\text{LaAlO}_3$ phosphors have been studied by SEM using backscattered electron detection mode. The black and white SEM micrograph of the Er^{3+} -doped LaAlO_3 [Fig. 2(a)] and $\text{Er}^{3+}/\text{Yb}^{3+}:\text{LaAlO}_3$ [Fig. 2(b)] powders shows that crystallites with no uniform shapes and sizes of the order of microns. This non-uniformity of shape and size can be assigned to the non-uniform distribution of temperature and mass flow in the combustion process. For EDX analysis, entire area of the black and white SEM micrographs was analyzed with EDX mapping. The EDX mapping measurements were carried out for Er^{3+} -doped and $\text{Er}^{3+}/\text{Yb}^{3+}$ co-doped LaAlO_3 phosphors to analyze the composition of the clustered particles. EDX elemental mappings (colored images) show the uniform distribution of constituent elements, which indicates a homogeneous distribution of each element in the powders and dopant ions are well dispersed even at the atomic level. Fig. 3(a) and (b) shows the EDX spectra of Er^{3+} doped LaAlO_3 and $\text{Er}^{3+}/\text{Yb}^{3+}$ co-doped LaAlO_3 phosphors, respectively. The EDX spectrum demonstrates the presence of La, Al, O, Er and Yb elements in Er^{3+} doped and $\text{Er}^{3+}/\text{Yb}^{3+}$ co-doped LaAlO_3 phosphors.

3.3. FTIR Spectrum: Vibrational Properties

The optical and luminescence properties of optically active Er^{3+} and Yb^{3+} ions depend upon vibrational energies of the LaAlO_3 crystallites. These vibrational energies play vital role in de-excitations of optically active ions through multiphonon relaxation. To study the energy of the vibrational modes of LaAlO_3 , FTIR spectra have been recorded for

(a)



(b)

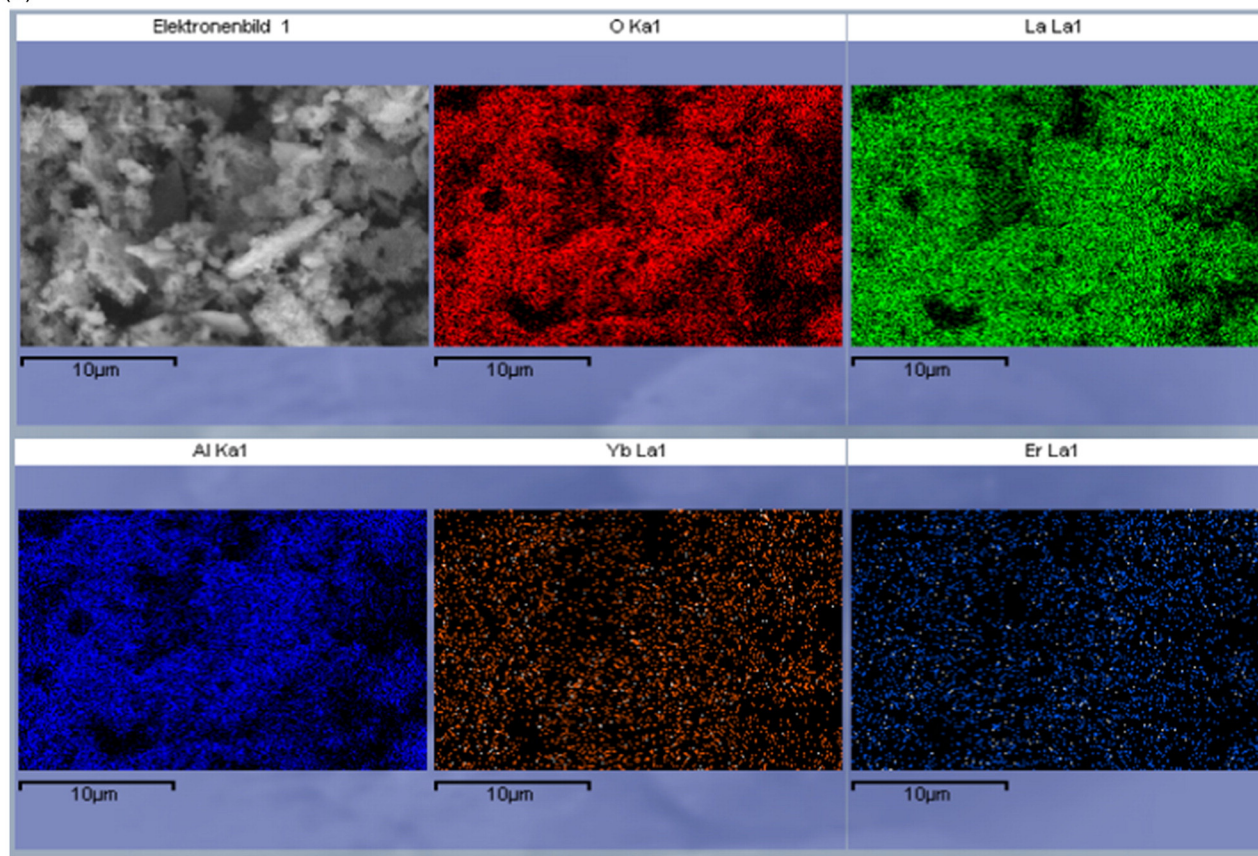


Fig. 2. EDX mapping of (a) $\text{LaAlO}_3:\text{Er}^{3+}$ and (b) $\text{Er}^{3+}/\text{Yb}^{3+}:\text{LaAlO}_3$ phosphors.

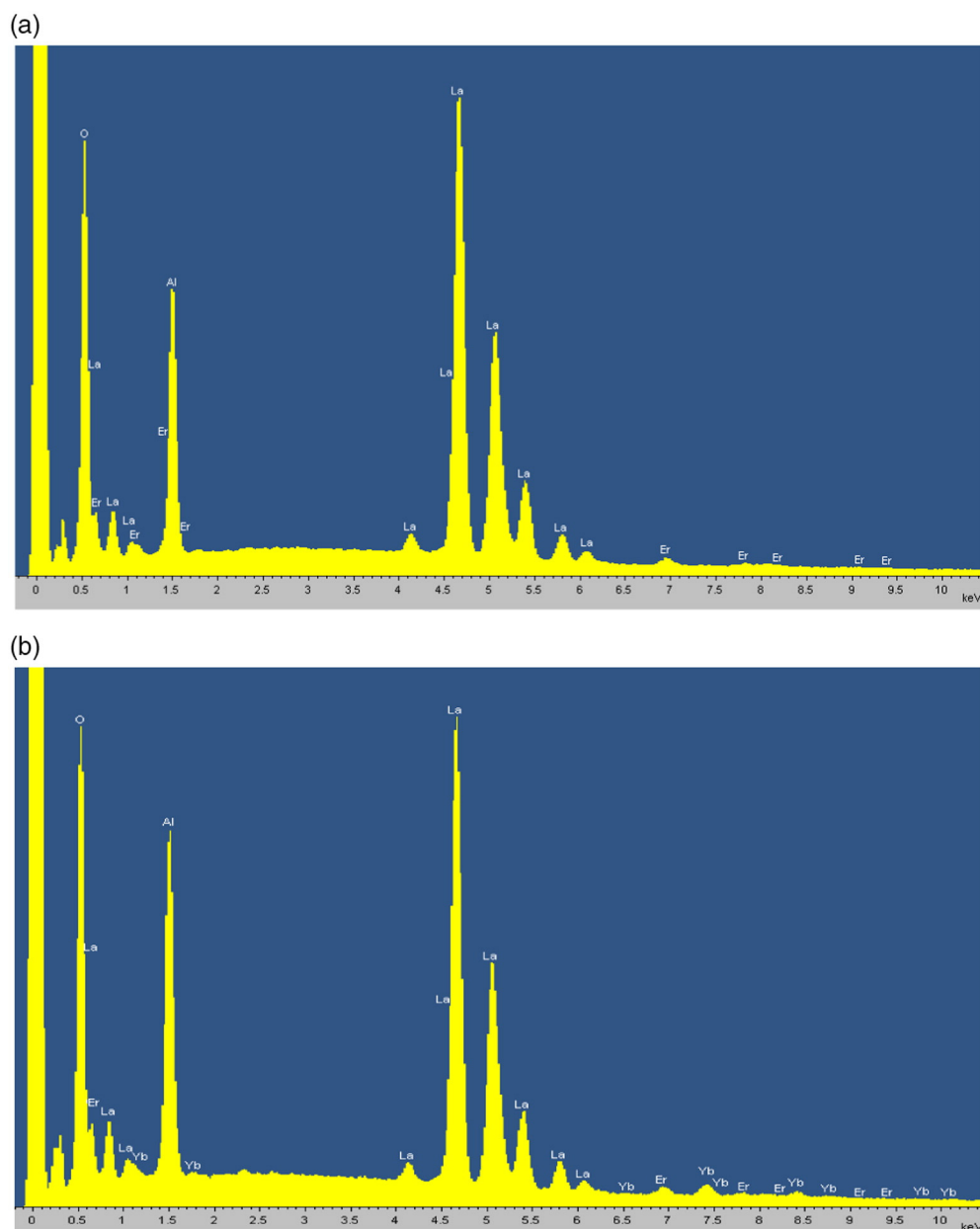


Fig. 3. EDX spectra of (a) $\text{LaAlO}_3:\text{Er}^{3+}$ and (b) $\text{Er}^{3+}/\text{Yb}^{3+}:\text{LaAlO}_3$ phosphors.

$\text{Er}^{3+}:\text{LaAlO}_3$ and $\text{Er}^{3+}/\text{Yb}^{3+}:\text{LaAlO}_3$ phosphors as shown in Fig. 4. The vibrational bands observed at $\sim 458\text{ cm}^{-1}$ and $\sim 668\text{ cm}^{-1}$ are due to the vibration of M-O (Al-O and La-O stretching frequencies) bonds showing the formation of rhombohedral phase of LaAlO_3 [18,23]. The peak around 1500 cm^{-1} may be due to hydroxyl group, assigned as bending modes of hydroxyl (OH) groups [22]. The low intensity vibrational bands at 2350 cm^{-1} and 2900 cm^{-1} correspond to asymmetric stretching of CO_2 and C-H groups, respectively. The broad peak centered around 3450 cm^{-1} corresponds to the O-H symmetric stretching from the surface hydroxyl group [23,24]. These surface groups enhance the depopulation of excited states of the Ln^{3+} ion so that they may act as a sink [22]. The hydroxyl groups provide an effective pathway for the radiationless energy transfer and quench the emission intensity. In case of $\text{Er}^{3+}/\text{Yb}^{3+}:\text{LaAlO}_3$ powder, the intensity of OH vibrational band is decreased due to presence of the Yb^{3+} ions which indicate the reduction in non-radiative processes in the co-doped phosphors.

3.4. Optical Properties

3.4.1. Absorption Spectra

The UV-Vis-NIR absorption spectrum of the Er^{3+} doped and the $\text{Er}^{3+}-\text{Yb}^{3+}$ co-doped LaAlO_3 phosphors in the 300–1800 nm region is shown in Fig. 5 (a) and (b). Ten absorption bands have been observed in the Er^{3+} doped phosphor peaking at $\sim 380\text{ nm}$, $\sim 409\text{ nm}$, $\sim 454\text{ nm}$, $\sim 490\text{ nm}$, $\sim 524\text{ nm}$, $\sim 545\text{ nm}$, $\sim 655\text{ nm}$, $\sim 796\text{ nm}$, $\sim 972\text{ nm}$ and $\sim 1494\text{ nm}$ corresponding to the transitions from the $^4\text{I}_{15/2}$ ground state to the $^4\text{G}_{11/2}$, $^2\text{H}_{9/2}$, $^4\text{F}_{3/2}$, $^4\text{F}_{5/2}$, $^2\text{H}_{11/2}$, $^4\text{S}_{3/2}$, $^4\text{F}_{9/2}$, $^4\text{I}_{9/2}$, $^4\text{I}_{11/2}$ and $^4\text{I}_{13/2}$ excited states of Er^{3+} ion. All the excited states. All the transitions are assumed to be electric dipole in nature, except the $^4\text{I}_{15/2} \rightarrow ^4\text{I}_{13/2}$ transition that shows a magnetic dipole contribution [25]. The energy levels positions of the different transitions of the Er^{3+} ion in the LaAlO_3 have been assigned according to the well-known Dieke's diagram for Er^{3+} ion in LaCl_3 crystal [26]. In the $\text{Er}^{3+}/\text{Yb}^{3+}:\text{LaAlO}_3$

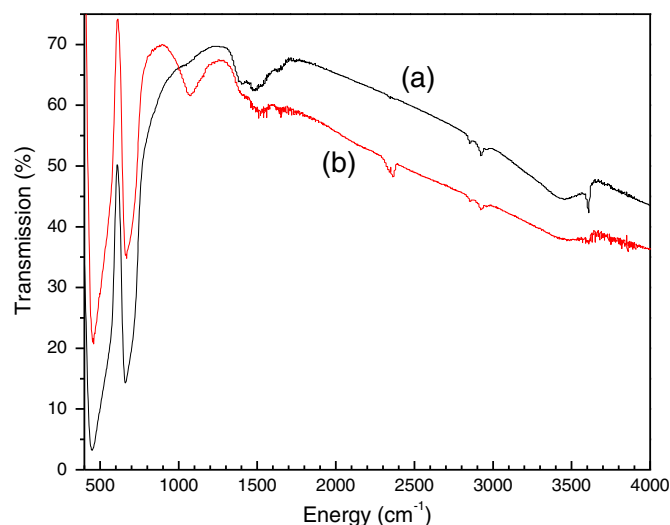


Fig. 4. Fourier transform infrared spectra of (a) $\text{Er}^{3+}:\text{LaAlO}_3$ and (b) $\text{Er}^{3+}/\text{Yb}^{3+}:\text{LaAlO}_3$ phosphors.

phosphor, the peak around ~ 972 nm appears broad due to the superposition of two absorption peaks corresponding to the $^2\text{F}_{7/2} \rightarrow ^2\text{F}_{5/2}$ and $^4\text{I}_{15/2} \rightarrow ^4\text{I}_{11/2}$ transitions of Yb^{3+} and Er^{3+} ions, respectively, which can be seen in Fig. 5(b). The increased absorption coefficient and broad absorption band of the $\text{Er}^{3+}/\text{Yb}^{3+}:\text{LaAlO}_3$ phosphor is relatively appropriate for the low cost 978 nm diode laser as a pump source. The high absorption cross-section around ~ 972 nm and good spectral overlap between the $^2\text{F}_{5/2} \rightarrow ^2\text{F}_{7/2}$ (Yb^{3+}) emission and the $^4\text{I}_{15/2} \rightarrow ^4\text{I}_{11/2}$ (Er^{3+}) absorption leads to an efficient ET from the Yb^{3+} to Er^{3+} in co-doped phosphor. The other absorption band position remains the same in both the phosphors.

3.4.2. Upconversion Luminescence in $\text{Er}^{3+}:\text{LaAlO}_3$ Phosphor

The upconversion (UC) emission spectrum of $\text{Er}^{3+}:\text{LaAlO}_3$ phosphor upon excitation with a laser radiation at 978 nm shows four upconversion emission bands in the 350–750 nm region corresponding

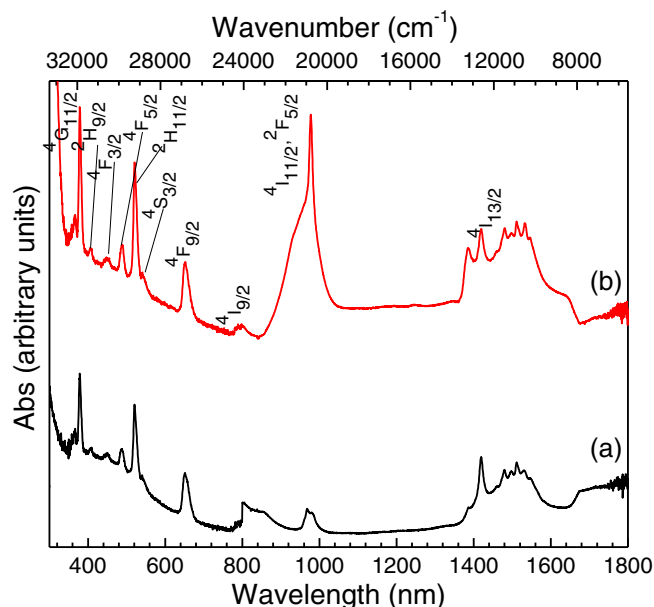


Fig. 5. Absorption spectra of (a) $\text{Er}^{3+}:\text{LaAlO}_3$ and (b) $\text{Er}^{3+}/\text{Yb}^{3+}:\text{LaAlO}_3$ phosphors.

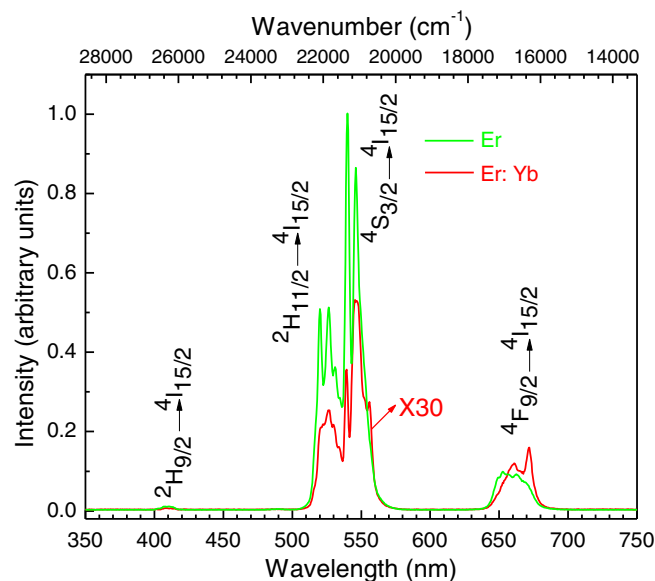


Fig. 6. Upconversion luminescence spectra of $\text{Er}^{3+}:\text{LaAlO}_3$ and $\text{Er}^{3+}/\text{Yb}^{3+}:\text{LaAlO}_3$ under 978 nm laser excitation.

to the $^2\text{H}_{9/2} \rightarrow ^4\text{I}_{15/2}$, $^2\text{H}_{11/2} \rightarrow ^4\text{I}_{15/2}$, $^4\text{S}_{3/2} \rightarrow ^4\text{I}_{15/2}$ and $^4\text{F}_{9/2} \rightarrow ^4\text{I}_{15/2}$ transitions peaking at ~ 408 nm, ~ 526 nm, 545 nm and 670 nm, respectively (Fig. 6). In order to know the number of NIR laser photons involved in the emission process, log-log plot of the photoluminescence intensity versus laser power has been made. The slopes are found to be 0.47, 1.29 and 1.1 for blue, green and red emissions, respectively. The lower slope values indicate that there is some effect of saturation on up-conversion intensities as the excitation power increases [27]. The UC emission peaking at ~ 408 nm should be due to the contribution of three NIR photons, whereas the green and red UC emission should be due to the involvement of two NIR photons [21]. The processes responsible for these UC emissions are caused by excited state absorption (ESA) [22,28] (Fig. 7). The NIR photons absorbed by the ground state Er^{3+} ions are directly excited to the $^4\text{I}_{11/2}$ state. A part of the excited Er^{3+} ions decays non-radiatively to the $^4\text{I}_{13/2}$ level and the rest part is promoted to the $^4\text{F}_{7/2}$ level by ESA. Then ($^2\text{H}_{11/2}$, $^4\text{S}_{3/2}$) thermalised levels are populated through non-radiative decay. Again ESA mechanism leads the Er^{3+} ions into the above $^4\text{G}_{11/2}$ level, then decays non-

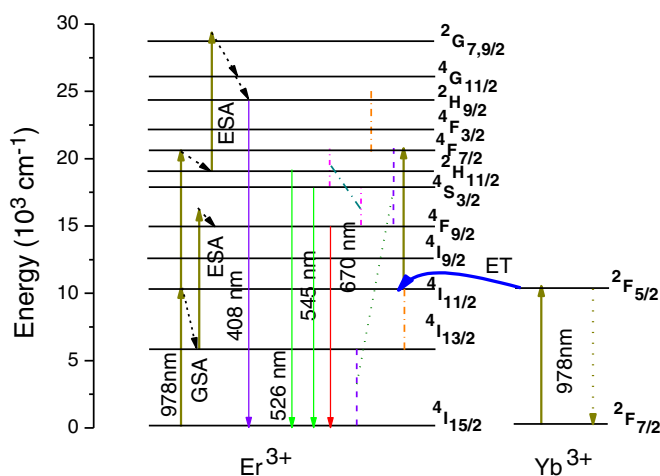


Fig. 7. Partial energy level structure of the Er^{3+} and Yb^{3+} for elucidation of the process that is responsible for the blue, green and red up-conversion emissions. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

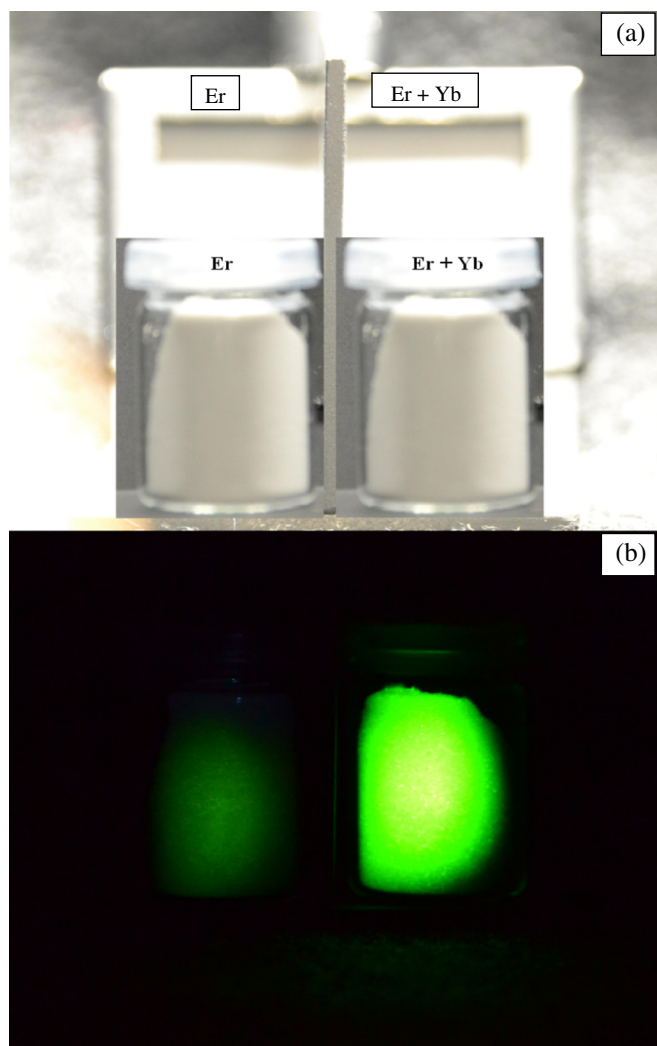


Fig. 8. Typical photographs of $\text{Er}^{3+}:\text{LaAlO}_3$ and $\text{Er}^{3+}/\text{Yb}^{3+}:\text{LaAlO}_3$ phosphors under (a) room light (appearance: a white powder) and (b) 978 nm laser light (appearance: green emission). The optical filter used to block the infrared transmitted laser light. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

radiatively to $^2\text{H}_{9/2}$ level. The weak blue emission is due to the $^2\text{H}_{9/2} \rightarrow ^4\text{I}_{15/2}$ radiative transition. The $^2\text{H}_{11/2}$ and $^4\text{S}_{3/2}$ levels are populated by the non-radiative relaxation from the $^4\text{F}_{7/2}$ level and then decay radiatively to the $^4\text{I}_{15/2}$ level with green emission. The $^4\text{F}_{9/2}$ level populated via the second photon absorption from the $^4\text{I}_{13/2}$ level decays radiatively and gives emission in the red region corresponding to the $^4\text{F}_{9/2} \rightarrow ^4\text{I}_{15/2}$ transition.

3.4.3. Effect of Yb^{3+} as Co-Dopant on Upconversion Luminescence of $\text{Er}^{3+}:-\text{LaAlO}_3$ Phosphor

The UC emission spectrum of the $\text{Er}^{3+}:\text{LaAlO}_3$ phosphor co-doped with Yb^{3+} ions upon direct excitation with 978 nm diode laser is shown in Fig. 6. Similar to $\text{Er}^{3+}:\text{LaAlO}_3$ phosphors blue, green, and red up-conversion emissions are observed but with enhanced intensity. The upconversion emission bands lying in the blue, green and red regions corresponding to the $^2\text{H}_{9/2} \rightarrow ^4\text{I}_{15/2}$, $^2\text{H}_{11/2} \rightarrow ^4\text{I}_{15/2}$, $^4\text{S}_{3/2} \rightarrow ^4\text{I}_{15/2}$ and $^4\text{F}_{9/2} \rightarrow ^4\text{I}_{15/2}$ transitions has been improved by a factor of about ~20, ~43, ~54 and ~22 times compared to the singly doped phosphor. This enhancement is due to the efficient energy transfer from the Yb^{3+} ions to the Er^{3+} ions. Similar kind of result was obtained in $\text{Er}^{3+}/\text{Yb}^{3+}$ co-doped Y_2O_3 microspheres by Guo et al. [29]. The log-log plots of power dependent up-conversion spectra in blue, green and red regions

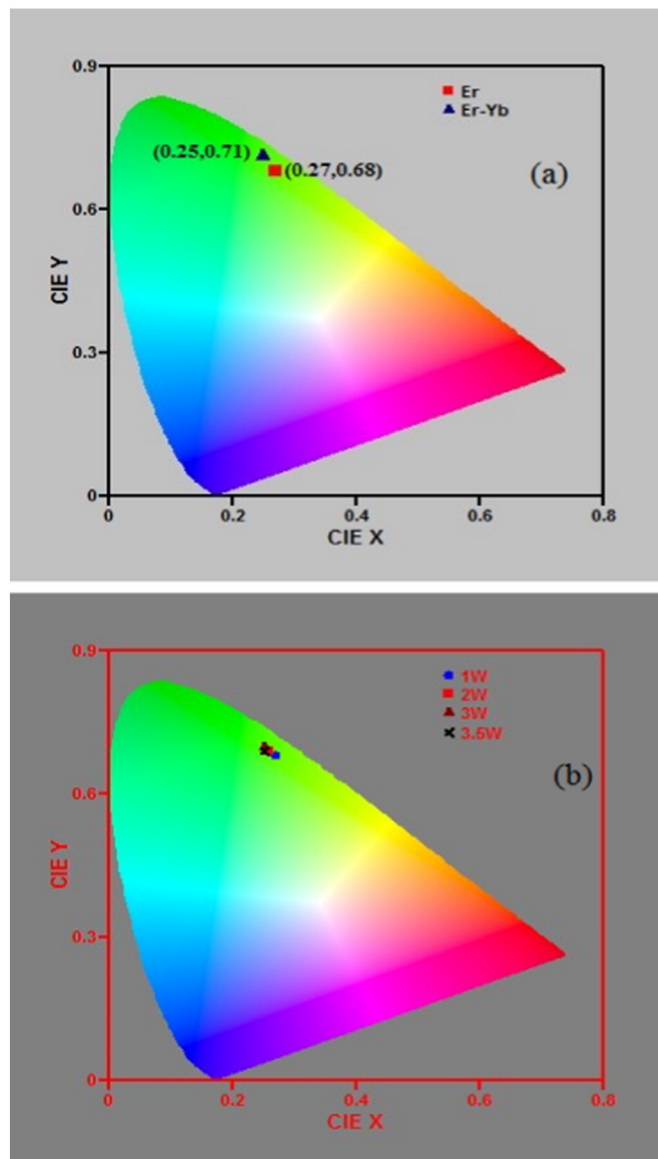


Fig. 9. Chromaticity diagram to visualize the color tunability under excitation with a diode laser at ~978 nm (a) $\text{Er}^{3+}:\text{LaAlO}_3$ and $\text{Er}^{3+}/\text{Yb}^{3+}:\text{LaAlO}_3$ phosphor (b) $\text{Er}^{3+}/\text{Yb}^{3+}:\text{LaAlO}_3$ phosphor at different pump power. The energy value of the diode laser at 978 nm is $\sim 10,224 \text{ cm}^{-1}$.

of Er^{3+} doped and $\text{Er}^{3+}/\text{Yb}^{3+}$ co-doped LaAlO_3 phosphors have been made. The slopes are found to be 0.05, 1.36 and 0.99 for blue, green and red emissions, respectively. The lower slope values indicate that there is some effect of saturation [27]. The phonon energy of the prepared phosphor ($\sim 458 \text{ cm}^{-1}$) is smaller than the other host materials [30–32]. Therefore, the Yb^{3+} ions from its ground state after receiving efficiently the photon energy ($\sim 10,224 \text{ cm}^{-1}$) of ~978 nm radiation due to the larger oscillator strength compared to that of Er^{3+} ion from the diode laser are pumped to the $^2\text{F}_{5/2}$ excited state, from where the Yb^{3+} ions relax to the ground state by transferring their excitation energy to its neighboring Er^{3+} ion. In the co-doped phosphor the energy of the $^4\text{I}_{11/2}$ state of Er^{3+} is very similar to the $^2\text{F}_{5/2}$ state of Yb^{3+} . Therefore, excited Yb^{3+} ions in the $^2\text{F}_{5/2}$ state efficiently transfer their excitation energy to the Er^{3+} ions in the ground state and promote them to the $^4\text{I}_{11/2}$ state. Again a part of the population in the $^4\text{I}_{11/2}$ state through the energy transfer from the excited Yb^{3+} ions is pumped to the next higher $^4\text{F}_{7/2}$ state of Er^{3+} . Some of the Er^{3+} ions are excited to above $^4\text{G}_{11/2}$ level through energy transfer from excited Yb^{3+} ions and then they decay non-radiatively to $^2\text{H}_{9/2}$ level. The blue emission is due to

$^2H_{9/2} \rightarrow ^4I_{15/2}$ radiative transition and its enhancement in co-doped one is apparently due to the cross-relaxation ($^4I_{11/2}; ^4F_{7/2} \rightarrow ^4I_{13/2}; ^2H_{9/2}$) energy transfer process. Excited Er^{3+} ions in the $^4F_{7/2}$ state decay non-radiatively to the ($^2H_{11/2}, ^4S_{3/2}$) and $^4F_{9/2}$ states from where green and red emission, respectively, is observed due to ($^2H_{11/2}, ^4S_{3/2} \rightarrow ^4I_{15/2}$ and $^4F_{9/2} \rightarrow ^4I_{15/2}$ transition as described before for Er^{3+} -doped system. The red emission may also be due to another channel to populate $^4F_{9/2}$ level and is explained as follows. The $^4I_{13/2}$ level may be populated due to the non-radiative relaxation from the $^4I_{11/2}$ level via energy transfer from the excited Yb^{3+} ions in the $^2F_{5/2}$ level is again promoted to the $^4F_{9/2}$ level. The Er^{3+} ions in the $^4F_{9/2}$ level relax radiatively to the ground $^4I_{13/2}$ level and gives photon in the red region corresponding to the $^4F_{9/2} \rightarrow ^4I_{15/2}$ transition.

The co-doping with Yb^{3+} in the doped phosphor modifies the ratio of green/red emission bands from about ~3.64 (for $Er^{3+}:\text{LaAlO}_3$) to ~6.96 (for $Er^{3+}/Yb^{3+}:\text{LaAlO}_3$) respectively. This is clearly observed in Fig. 8 where enhancement in green emission has been observed for $Er^{3+}/Yb^{3+}:\text{LaAlO}_3$ when compare with $Er^{3+}:\text{LaAlO}_3$. Due to the presence of the Yb^{3+} ions in the co-doped phosphor the distance between Ln^{3+} ions in the excited states is decreased which makes the interaction stronger between Ln^{3+} ions. This significant variation in the intensity ratio (about two times) in the co-doped $Er^{3+}/Yb^{3+}:\text{LaAlO}_3$ phosphor is due to the contributions of the processes involved through the ESA and efficient energy transfer (ET) from Yb^{3+} ions. As the intensity of the UC emission band corresponding to the $^4S_{3/2} \rightarrow ^4I_{15/2}$ transition remains always higher in both the doped/co-doped phosphor. This indicates that along with ESA/ET some other process namely the cross-relaxation energy transfer ($(^4I_{13/2}; ^4F_{9/2} \rightarrow ^4I_{15/2}; ^4F_{7/2})$ and $(^4F_{7/2}; ^4F_{9/2} \rightarrow ^4S_{3/2}; ^4S_{3/2})$) is also involved. This fact also suggests why the enhancement for green emission band is larger than the red emission band. The color emitted from the doped/co-doped phosphor before and after excitation at 978 nm radiation is shown in Fig. 8 (a) and (b).

Moreover, the enhancement in UC emission intensity corresponding to the $^2H_{11/2} \rightarrow ^4I_{15/2}$ transition is more compared to the $^4S_{3/2} \rightarrow ^4I_{15/2}$ and other transitions in the co-doped phosphor. The pump power dependence on UC emission intensity for both the bands have been done and their intensity ratio ($I_{526\text{ nm}}/I_{545\text{ nm}}$) has been monitored and noted to vary from 0.43 to 0.67 when the pump power of the diode laser lasing at ~978 nm (excitation energy ~10,224 cm^{-1}) is increased from 1.0 W to 3.5 W. This increase in the intensity ratio is basically due to the thermalization involved between the two closely spaced ($^2H_{11/2}$ and $^4S_{3/2}$) levels [33]. Thus, the large enhancement corresponding to the $^2H_{11/2} \rightarrow ^4I_{15/2}$ transition is also supported by the thermalization involved between the $^2H_{11/2}$ and $^4S_{3/2}$ levels along with the aforementioned processes. The upconversion emission intensity was found to increase with increase in the excitation power and hence the population densities in the upper states. The color of radiation emitted from the doped/co-doped phosphor is not tuned significantly with increase in the pump power whereas it is tuned towards the pure green side with the incorporation of the Yb^{3+} ion (Fig. 9). The color coordinates are found to be (0.29, 0.68), (0.28, 0.68) and (0.27, 0.68), respectively, for 1 W, 2 W, 3 W, and 3.5 W pump powers. The small mismatch in the color coordinates with increase in the pump power may be due to the fluctuations in the pump power as well as in the alignments during the different scans. The color tunability in the phosphor co-doped with Yb^{3+} ions upon excitation at 978 nm and at different excitation pump power may be visualized by the chromaticity diagram (Fig. 9). This shows the potentiality of present phosphor to be used in making the NIR to visible upconverter and green display devices.

4. Conclusions

The Er^{3+} doped and Er^{3+}/Yb^{3+} co-doped LaAlO_3 phosphors were successfully prepared by combustion method. The XRD result indicates that the influence of doping and co-doping is not evident on the crystal structure. From morphological and elemental analysis it was observed

that phosphors show the crystallites with no uniform shapes and sizes of the order of microns. The EDX elemental mappings show the uniform distribution of constituent elements, which indicates a homogeneous distribution of each element in the phosphors and dopant ions are well dispersed even at the atomic level. The vibrational energies of Er^{3+} doped and Er^{3+}/Yb^{3+} co-doped LaAlO_3 phosphors have been analyzed using FTIR spectra. The intense up-conversion emission has been observed in $Er^{3+}:\text{LaAlO}_3$ and $Er^{3+}/Yb^{3+}:\text{LaAlO}_3$ phosphors. The red to green up-conversion intensity ratio is increased in Er^{3+}/Yb^{3+} co-doped LaAlO_3 phosphors. The ESA mechanism was responsible to observe the up-converted luminescence in Er^{3+} doped LaAlO_3 phosphors, whereas ETU and cross relaxation channels are responsible to observe up-conversion luminescence in the Er^{3+}/Yb^{3+} co-doped LaAlO_3 phosphors. The color of radiation emitted from the doped/co-doped phosphor is not tuned significantly with increase in the pump power where as it is tuned towards the pure green side with the incorporation of the Yb^{3+} ion. All these results are a sign of that these materials could be suitable for up-conversion lasers and green display device applications.

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